

Intro to Frequency Response

EGR 345

GVSU

Outline

- some thoughts on the projects
 - additional thoughts on on building vibration absorbers
- response of MSD system to sinusoidal input at various frequencies

Initial Project Thoughts

- things are really coming together
 - with your help

Building Vibration Absorbers

- step 1: building without absorber (no pendulum)
- step 2: passive pendulum
- step 3: magnetic pendulum damping
- step 4: active pendulum
 - pendulum attached to DC motor or servo
 - feedback control to create motor/servo input

What would Dr. Farris say about our projects?

- I think there is something he harps on
- I think there is a false dichotomy between ME and PDM

Analysis of building vibration absorbers

- what's wrong with trial and error?
 - I think I could make an argument from every major in the room

How do passive vibration absorbers work?

- and what role does damping play?

Intro to Frequency Response

How does an MSD system response to a sinusoidal input?

- low frequency, medium frequency, and high frequency

